Quiz 3 (Version 4)

CS 320: Concepts of Programming Languages

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- ▷ You will have approximately 35 minutes to complete this quiz. Make sure to read every question; some are easier than others.
- ▷ Make it very clear what your final solution for each problem is (e.g., by surrounding it in a box). We reserve the right to mark off points if we cannot tell what your final solution is.
- ▷ You must show your work on all problems unless otherwise specified. A solution without work will be considered incorrect (and will be investigated for potential academic misconduct).

1 OCaml Programming

(10 points) Implement the function

```
val filter2_opt : ('a -> 'b -> bool) -> 'a list -> 'b list -> ('a * 'b) list option
```

so that `filter2_opt p l r` is `Some` of the list of pairs `(x, y)` of corresponding elements of `l` and `r` such that `p x y` is `true`, in order. If `l` and `r` have different lengths, then the result is `None`.

```
let _ = assert(filter2_opt (<) [1;5;3;2] [4;2;6;2] = Some [(1,4);(3,6)])
let _ = assert(filter2_opt (<) [1;2;3] [4;5] = None)
let _ = assert(filter2_opt (<) [3;4] [1;2] = Some [])
let _ = assert(filter2_opt (<) [] [] = Some [])

let p s n = String.length s = n
let _ = assert(filter2_opt p ["a";"bc";"def"] [1;3;3] = Some [("a",1);("def",3)])
```

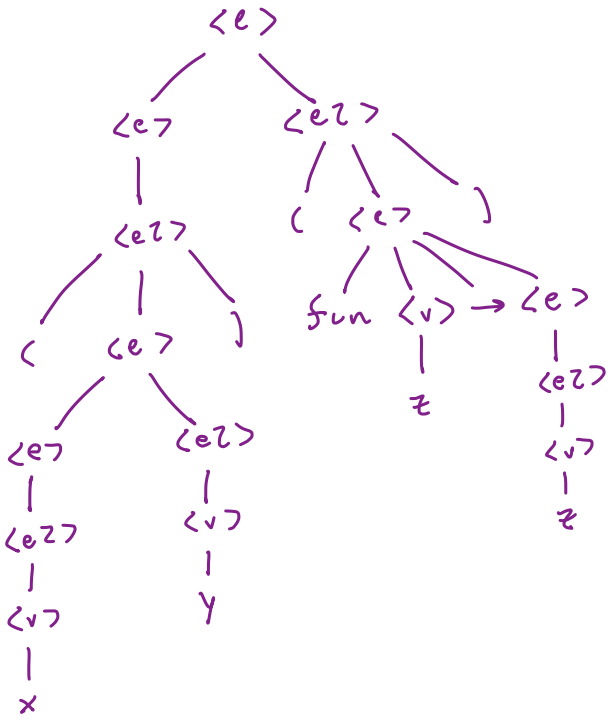
```
let filter2_opt p l r =
  let rec go l r =
    match l, r with
    | [], [] -> Some []
    | [], _ -> None
    | _, [] -> None
    | x::l, y::r ->
      match go l r with
      | Some s ->
        (if f x y then [x,y] else [])@s
      | None -> None
  in go l r
```

2 Formal Grammar

Consider the following BNF grammar.

```
<e>  ::= fun <v> -> <e> | <e> <e2> | <e2>
<e2> ::= <v> | (<e>)
<v>  ::= x | y | z
```

A. (10 points) Demonstrate that the sentence `(x y) (fun z -> z)` is recognized by the above grammar by providing a parse tree for it.



The grammar from the previous part is reproduced for convenience.

```
<e>  ::= fun <v> -> <e> | <e> <e2> | <e2>
<e2> ::= <v> | (<e>)
<v>  ::= x | y | z
```

B. (10 points) Demonstrate that the above grammar is ambiguous by determining a sentence with multiple parse trees. Include both parse trees in your answer.

fun x → x x

